AI.211

Pattern Recognitio

COURSE OVERVIEW

Classifying and clustering patterns: feature extraction, classifiers, and evaluation for perception and diagnostics.



ILLUSTRATIVE HERO IMAGERY

PATTERN RECOGNITION

1

SESSION

60 MINUTES

ORIENTATION: SEEING PATTERNS

KEY TOPICS

- What is pattern recognition
- Feature extraction and selection
- Supervised vs. unsupervised
- Applications

TEACHING FOCUS

Recognize structure in data.

KEY TAKEAWAY Patterns are the signature of meaning.

PATTERN RECOGNITION

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SESSION

60 MINUTES

CLASSICAL CLASSIFIERS

KEY TOPICS

- Nearest neighbor
- Linear discriminant analysis
- Naive Bayes
- Support vector machines

TEACHING FOCUS

Assign patterns to categories.

KEY TAKEAWAY Classifiers draw boundaries in feature space.

3

SESSION

60 MINUTES

FEATURE ENGINEERING

KEY TOPICS

- Descriptors and histograms
- Dimensionality reduction
- Feature selection
- Domain-specific features

TEACHING FOCUS

Represent patterns compactly.

KEY TAKEAWAY Good features make classification easy.

PATTERN RECOGNITION

4

SESSION

60 MINUTES

CLUSTERING

KEY TOPICS

- k-means and variants
- Gaussian mixture models
- DBSCAN
- Cluster evaluation

TEACHING FOCUS

Group similar patterns together.

KEY TAKEAWAY Clustering finds categories without labels.

PATTERN RECOGNITION

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SESSION

60 MINUTES

EVALUATION

KEY TOPICS

- Confusion matrix
- ROC and AUC
- Cross-validation
- Precision-recall

TEACHING FOCUS

Judge a recognizer honestly.

KEY TAKEAWAY Accuracy alone can mislead.

PATTERN RECOGNITION

6

SESSION

60 MINUTES

PATTERN RECOGNITION IN AI & REVIEW

KEY TOPICS

- Speech, vision, and sensor patterns
- Course recap and assessment
- Toward robot kinematics (AI.212)
- Recognizer demo

TEACHING FOCUS

Build a pattern recognizer end-to-end.

KEY TAKEAWAY Recognition turns raw data into categories.